

1] Encrypt 'CHASE YOUR DREAMS' using
a. Caesar cipher

$$\Rightarrow n = 3$$

$$C = (P + 3) \bmod 26$$

Plain text	$P + 3$	$(P + 3) \bmod 26$	Ciphertext
C	5	5	F
H	10	10	K
A	3	3	D
S	21	21	V
E	7	7	H
Y	27	1	B
O	17	17	R
U	23	23	X
R	20	20	U
D	6	6	G
R	20	20	U
E	7	7	H
A	3	3	D
M	15	15	P
S	21	21	V

Ciphertext : FKDVH BRXU GUHDPV

b. Shift cipher with key 7

$$C = (P+7) \bmod 26$$

Plaintext	P+7	$(P+7) \bmod 26$	Ciphertext
C	9	9	J
H	14	14	O
A	7	7	H
S	25	25	Z
E	11	11	L
Y	31	5	F
O	21	21	V
U	27	1	B
R	24	24	Y
D	10	10	K
R	24	24	Y
E	11	11	L
A	7	7	H
M	19	19	T
S	25	25	Z

Ciphertext: JOHZL FVBY KYLHTZ

c. ROT 13

Plaintext	$P+13$	$(P+13) \bmod 26$	Ciphertext
C	15	15	P
H	20	20	U
A	13	13	N
S	31	5	F
E	17	17	R
Y	31	11	L
O	27	1	B
U	33	7	H
R	30	4	E
D	16	16	Q
R	30	4	E
E	17	17	R
A	13	13	N
M	25	25	Z
S	31	5	F

Ciphertext = PUNFR LBHE QERNZF

2] Decrypt

a. "WRWDOOB VHFUHW PHIVVDJH Using Caesar cipher

Ciphertext	$P(C-3)$	$(C-3) \bmod 26$	Plaintext
W=22	19	19	T
R=17	14	14	O
W=22	19	19	T
D=3	0	0	A
O=14	11	11	L
O=14	11	11	L
B=1	-2	24	Y
V=21	18	18	S
H=7	4	4	E
F=5	2	2	C
U=20	17	17	R
H=7	4	4	E
W=22	19	19	T
P=15	12	12	M
H=7	4	14	E
V=21	18	18	S
V=21	18	18	S
D=3	0	10	A
J=9	6	6	G
H=7	4	4	E

Plaintext = TOTALLY SECRET MESSAGE

6. MJQQT BTWQT Shift Cipher 5

Ciphertext	C-5	(C-5) mod 26	Plaintext
M = 12	7	7	H
J = 9	4	4	E
Q = 16	11	11	L
Q = 16	11	11	L
T = 19	14	14	O
B = 1	-4	22	W
T = 19	14	14	O
W = 22	17	17	R
Q = 16	11	11	L
I = 8	3	3	D

Plaintext = HELLO WORLD

SJB DTWP YNRJ shift Cipher S

Ciphertext C-5 (C-5) mod 26 Plaintext

S = 18 13 13 N

J = 9 4 4 E

B = 1 -4 22 W

D = 3 -2 24 Y

T = 19 14 14 O

W = 22 17 17 R

P = 15 10 10 K

Y = 24 19 19 T

N = 13 8 8 I

R = 17 12 12 M

J = 9 4 4 E

Plaintext = NEW YORK TIME

C. "FRAG ZBER PBSSRR" using ROT13

Ciphertext	C-13	$(C-13) \bmod 26$	Plaintext
F = 5	-8	18	S
R = 17	4	4	E
A = 0	-13	13	N
G = 16	3	3	D
Z = 25	12	12	M
B = 1	-12	14	O
E = 4	-9	17	R
R = 17	4	4	E
P = 15	2	2	C
B = 1	-12	14	O
S = 18	5	5	F
S = 18	5	5	F
R = 17	4	4	E
R = 17	4	4	E

Plaintext = SEND MORE COFFEE